

1. **A new compound developed in the lab was tested as a mutagen. When the Ames test was performed, the results showed the presence of many bacterial colonies following exposure. This result means the chemical \_\_\_\_\_.**
  - ☐ a. causes apoptosis.
  - ☐ b. is radioactive.
  - ☐ c. would be a great candidate to use for chemotherapy.
  - ☐ d. is a mutagen and also may be a carcinogen.
  - ☐ e. is safe.
2. **The viral oncogene associated with the Rous sarcoma virus is the \_\_\_\_\_.**
  - ☐ a. ERBB2 gene.
  - ☐ b. v-sis gene.
  - ☐ c. src gene.
  - ☐ d. p53 gene.
  - ☐ e. RAS gene.
3. **Which of the following is a lipophilic primary message that crosses the plasma membrane and interacts with a cytosolic receptor?**
  - ☐ a. Epidermal Growth Factor
  - ☐ b. TGF-beta
  - ☐ c. Estrogen
  - ☐ d. Pertussis toxin
  - ☐ e. PDGF
4. **Each of the following components is associated with the primary cell wall except \_\_\_\_\_:**
  - ☐ a. lignins.
  - ☐ b. cellulose microfibrils.
  - ☐ c. hemicelluloses.
  - ☐ d. pectins.
  - ☐ e. glycoproteins.
5. **Each of the following is true about Ras except \_\_\_\_\_.**
  - ☐ a. Ras activity is inhibited by GTPase activating proteins (GAPs).
  - ☐ b. Ras activates a cascade of phosphorylation events within the cell.

- c. Ras binds steroid hormones.
  - d. Ras is monomeric.
  - e. Ras is activated by Sos, a guanine nucleotide exchange factor.
6. **Each of the following is true about basal lamina except \_\_\_\_\_.**
- a. it is composed primarily of fibronectin.
  - b. it serves as a structural support.
  - c. it can influence cell migrations in certain areas of the body.
  - d. it allows for movement of some small molecules but blocks movement of large molecules.
7. **Crawling cells produce broad, thin protrusions known as \_\_\_\_\_ and pointed structures called \_\_\_\_\_ at the leading edge of the cell.**
- a. filopodia; lamellipodia
  - b. podosome; lamellipodia
  - c. pseudopodia; focal contacts
  - d. lamellipodia; filopodia
8. **Actin bundles give shape and support to \_\_\_\_\_ found in intestinal mucosal cells.**
- a. flagella
  - b. nuclei
  - c. microvilli
  - d. lamellipodia
  - e. cilia
9. **Compounds that inhibit receptors by preventing the natural messenger from binding are known as \_\_\_\_\_.**
- a. antagonists.
  - b. agonists.
  - c. syndecans.
  - d. arrestins.
10. **The promotion phase of carcinogenesis is enhanced by \_\_\_\_\_.**
- a. natural selection of cells exhibiting enhanced growth rate and invasive properties.
  - b. a single base-pair change in genomic DNA.

- c. inactivation of Ras activity.
- d. prolonged or repeated exposure to foreign or natural agents that stimulate cell proliferation.
- e. a single exposure to a carcinogen.

**Answers for Questions 1-10:**

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 |
|----|----|----|----|----|----|----|----|----|-----|
| D  | C  | C  | A  | C  | A  | D  | C  | A  | D   |

**Questions (Continued):**

**11. Elastin molecules are crosslinked to each other by \_\_\_\_.**

- a. peptide bonds between glycines.
- b. hydrophobic interactions between hydroxyprolines and glycines.
- c. none of these choices (above or below).
- d. hydrogen bonds between hydroxylysines and hydroxyprolines.
- e. covalent bonds between prolines.

**12. Cytoskeletal elements are found in \_\_\_\_.**

- a. all of these choices (above and below).
- b. archaea and eukaryotes.
- c. eukaryotes.
- d. bacteria.
- e. archaea.

**13. Each of the following is true about plant pectins except \_\_\_\_.**

- a. pectins trap and bind water molecules.
- b. pectins are long unbranched polymers.
- c. pectins form the matrix in which cellulose microfibrils are embedded.
- d. pectins have a gel-like consistency.
- e. pectins bind adjacent cell walls together.

14. Each of the following is an intracellular messenger except \_\_\_\_\_.

- ☐ a. calcium ions.
- ☐ b. cAMP.
- ☐ c. PDGF.
- ☐ d. DAG.
- ☐ e. IP3.

15. Each of the following is an example of microfilament-based motility and contractility except \_\_\_\_\_.

- ☐ a. cytoplasmic streaming.
- ☐ b. cytokinesis.
- ☐ c. trailing edge retraction in a moving cell.
- ☐ d. ciliary motility.
- ☐ e. pseudopodia formation.

16. Children with hereditary retinoblastoma inherit a chromosome 13 from one parent that has which of the following defects in the RB gene?

- ☐ a. duplication
- ☐ b. inversion
- ☐ c. insertion
- ☐ d. point mutation
- ☐ e. deletion

17. Transforming growth factor beta receptor is a \_\_\_\_\_.

- ☐ a. Receptor serine-threonine kinase
- ☐ b. Non receptor serine/threonine kinase
- ☐ c. G-protein coupled receptor
- ☐ d. Receptor tyrosine kinase

18. Which of the following would be correctly described as juxtacrine signaling?

- ☐ a. a signal that requires physical contact between cells
- ☐ b. a signal that acts on the cell that produces it
- ☐ c. a signal that diffuses over long distances from the producing cell

- d. a signal that diffuses over short distances from the producing cell
- e. a hormonal signal

19. **N-CAM allows interactions between neural cells and is a member of which of the following protein families?**

- a. lectins
- b. catenins
- c. selectins
- d. immunoglobulin superfamily (IgSF) proteins
- e. cadherins

20. **Microfilaments function in \_\_\_\_\_.**

- a. maintenance of animal cell shape.
- b. cell division.
- c. cytoplasmic streaming.
- d. all of these (above and below).
- e. muscle contraction.

**Answers for Questions 11-20:**

| Q11 | Q12 | Q13 | Q14 | Q15 | Q16 | Q17 | Q18 | Q19 | Q20 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| C   | A   | B   | C   | D   | E   | A   | A   | D   | D   |

**Questions (Continued):**

21. **Most hereditary forms of breast and ovarian cancer arise in women who inherit a mutant copy of either the BRCA1 or BRCA2 gene, each of which encodes proteins involved directly in \_\_\_\_\_.**

- a. inhibiting apoptosis.
- b. a spindle assembly checkpoint.
- c. repairing DNA double-strand breaks.
- d. passing through the G1 restriction point.
- e. activating Rho proteins to increase cell motility.

22. **Oncogenes encode proteins in each of the following categories except \_\_\_\_\_.**

- a. growth factors.
- b. nonreceptor protein kinases.
- c. cytochrome P450 enzymes.
- d. transcription factors.
- e. growth-factor receptors.

23. **In the axoneme, dynein sidearms are anchored on \_\_\_\_\_.**

- a. B tubules.
- b. interdouplet nexin.
- c. A tubules.
- d. central pair tubules.
- e. radial spokes.

24. **CapZ \_\_\_\_\_.**

- a. connects microfilaments to microtubules.
- b. causes microfilaments to lengthen by assembly at their plus ends.
- c. forms microfilament branches.
- d. depolymerizes microfilaments at their plus ends.
- e. prevents assembly at microfilament plus ends.

25. **Which of the following statements is correct with respect to flagella and cilia?**

- a. Cilia often have a helical pattern of movement.
- b. Cilia generally are longer than flagella.
- c. Cilia generate a force parallel to the cell surface.
- d. The flagellum is usually at the leading end of a swimming sperm cell.
- e. There usually are fewer cilia than flagella.

26. **Which of the following proteins attaches a crawling cell to the substrate in a focal adhesion?**

- a. integrin
- b. actin
- c. myosins
- d. alpha-actinin

27. The \_\_\_\_\_ is a complex network of interconnected filaments and tubules that extends throughout the cytosol from the nucleus to the inner surface of the plasma membrane.
- a. cytoskeleton
  - b. desmin sheath
  - c. microfilament array
  - d. axoneme
  - e. microtubule apparatus
28. Striated muscle has light and dark bands. The light bands are called \_\_\_\_\_, whereas the dark bands are called \_\_\_\_\_.
- a. H zones; M zones
  - b. Z lines; M lines
  - c. M lines; A bands
  - d. I bands; A bands
  - e. A bands; I bands
29. In cartilage, proteoglycans form large complexes by attaching to \_\_\_\_\_.
- a. elastin.
  - b. glycosaminoglycans.
  - c. laminin.
  - d. collagen.
  - e. hyaluronate.
30. How do carcinogens damage DNA?
- a. by causing breaks in one or both strands.
  - b. by generating cross-links between DNA strands.
  - c. all these choices (above and below) are mechanisms by which carcinogens can damage DNA.
  - d. by hydroxylating DNA bases.
  - e. by binding to DNA and directly disrupting base pairing.

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**Answers for Questions 21-30:**

| Q21 | Q22 | Q23 | Q24 | Q25 | Q26 | Q27 | Q28 | Q29 | Q30 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| C   | C   | C   | E   | D   | A   | A   | D   | E   | C   |

**Questions (Continued):**

**31. The expression of which of the following contributes to development of cancer?**

- ☐ a. oncogenes
- ☐ b. apoptosis genes
- ☐ c. proto-oncogenes
- ☐ d. tumor suppressor genes
- ☐ e. the p53 gene

**32. Dynamic assembly and disassembly of a cytoplasmic microtubule in a cell occurs primarily at its plus end, because its minus end is usually anchored to \_\_\_\_\_.**

- ☐ a. a centromere.
- ☐ b. a microtubule organizing center.
- ☐ c. the plasma membrane.
- ☐ d. a mitochondrial outer membrane.
- ☐ e. a basal body.

**33. IP3 receptors are associated with \_\_\_\_\_.**

- ☐ a. mitochondrial membranes.
- ☐ b. ER membranes.
- ☐ c. endosome membranes.
- ☐ d. plasma membranes.
- ☐ e. Golgi apparatus membranes.

**34. cAMP activates \_\_\_\_\_.**

- ☐ a. MAPK.
- ☐ b. protein kinase A.
- ☐ c. protein kinase G.
- ☐ d. protein kinase B.
- ☐ e. protein kinase C.

35. Which of the following forms the insoluble layer of aromatic alcohols found primarily in woody tissues?

- ☐ a. proteoglycans
- ☐ b. lignins
- ☐ c. cadherins
- ☐ d. pectins
- ☐ e. hyaluronates

36. Which of the following sequences is in the correct order for the process by which a myosin head generates force on an actin filament for muscle contraction?

- ☐ a. crossbridge formation, cocking of myosin head, power stroke, crossbridge dissociation
- ☐ b. power stroke, crossbridge formation, crossbridge dissociation, cocking of myosin head
- ☐ c. crossbridge formation, power stroke, crossbridge dissociation, cocking of myosin head
- ☐ d. cocking of myosin head, crossbridge formation, power stroke, crossbridge dissociation

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| Q31 | Q32 | Q33 | Q34 | Q35 | Q36 |
| A   | B   | B   | B   | B   | A   |